# GSI Discussion Session 02

Information Asymmetry

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# Asymmetric Information

- We start with a simple model
- True value V is either:
  - $V_L$ a lower price or,
  - $V_H$ a higher price, with equal probability, and $\sigma = V_H - V_L > 0$
- Two types of traders
  - Informed traders: Knows if V is $V_H$ or $V_L$
  - Uninformed traders: Buy/sell with equal prob.
  - There are $\alpha$ informed traders in the market

# How do the Traders Trade

- Informed Traders (they know the price…)
  - Buy if $V = V_H$, sell if $V = V_L$
- Uninformed traders (they have no idea…)
  - P(Buy)=0.5 regardless of price

# The Dealer

- The dealer sees trades in the market, and has to guess what V is
- $V^D_{t-1}=E[V|F_{t-1}]=\theta_{t-1}V_H+(1-\theta_{t-1})V_L$
- $F_{t-1}$ is the dealer's information (order flow)
- $\theta_{t-1}$ is the dealer's belief the price will be $V_H$ at time t

# Dealer's belief

- Dealer's believe that the asset is worth $V_H$ at time t is $\theta_{t-1}$
- This belief updates each time we see a trade coming in
  - $\theta_t(Buy) = P(V = V_H | x_t = Buy) = \frac{1+\alpha}{1+\alpha(2\theta_{t-1}-1)}\theta_{t-1} > \theta_{t-1}$
  - $\theta_t(Sell) = P(V = V_H | x_t = Sell) = \frac{(1-\alpha)\theta_{t-1}}{1+\alpha(1-2\theta_{t-1})}\theta_{t-1} < \theta_{t-1}$
- What does this look like?

Buy

| alpha \ theta | 0 | 0.1 | 0.2 | 0.3 | 0.4 | 0.5 | 0.6 | 0.7 | 0.8 | 0.9 | 1 |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 0 | 0.00 | 0.10 | 0.20 | 0.30 | 0.40 | 0.50 | 0.60 | 0.70 | 0.80 | 0.90 | 1.00 |
| 0.1 | 0.00 | 0.12 | 0.23 | 0.34 | 0.45 | 0.55 | 0.65 | 0.74 | 0.83 | 0.92 | 1.00 |
| 0.2 | 0.00 | 0.14 | 0.27 | 0.39 | 0.50 | 0.60 | 0.69 | 0.78 | 0.86 | 0.93 | 1.00 |
| 0.3 | 0.00 | 0.17 | 0.32 | 0.44 | 0.55 | 0.65 | 0.74 | 0.81 | 0.88 | 0.94 | 1.00 |
| 0.4 | 0.00 | 0.21 | 0.37 | 0.50 | 0.61 | 0.70 | 0.78 | 0.84 | 0.90 | 0.95 | 1.00 |
| 0.5 | 0.00 | 0.25 | 0.43 | 0.56 | 0.67 | 0.75 | 0.82 | 0.88 | 0.92 | 0.96 | 1.00 |
| 0.6 | 0.00 | 0.31 | 0.50 | 0.63 | 0.73 | 0.80 | 0.86 | 0.90 | 0.94 | 0.97 | 1.00 |
| 0.7 | 0.00 | 0.39 | 0.59 | 0.71 | 0.79 | 0.85 | 0.89 | 0.93 | 0.96 | 0.98 | 1.00 |
| 0.8 | 0.00 | 0.50 | 0.69 | 0.79 | 0.86 | 0.90 | 0.93 | 0.95 | 0.97 | 0.99 | 1.00 |
| 0.9 | 0.00 | 0.68 | 0.83 | 0.89 | 0.93 | 0.95 | 0.97 | 0.98 | 0.99 | 0.99 | 1.00 |
| 1 | | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 |

Sell

| alpha \ theta | 0 | 0.1 | 0.2 | 0.3 | 0.4 | 0.5 | 0.6 | 0.7 | 0.8 | 0.9 | 1 |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 0 | 0.00 | 0.01 | 0.04 | 0.09 | 0.16 | 0.25 | 0.36 | 0.49 | 0.64 | 0.81 | 1.00 |
| 0.1 | 0.00 | 0.01 | 0.03 | 0.08 | 0.14 | 0.23 | 0.33 | 0.46 | 0.61 | 0.79 | 1.00 |
| 0.2 | 0.00 | 0.01 | 0.03 | 0.07 | 0.12 | 0.20 | 0.30 | 0.43 | 0.58 | 0.77 | 1.00 |
| 0.3 | 0.00 | 0.01 | 0.02 | 0.06 | 0.11 | 0.18 | 0.27 | 0.39 | 0.55 | 0.75 | 1.00 |
| 0.4 | 0.00 | 0.00 | 0.02 | 0.05 | 0.09 | 0.15 | 0.23 | 0.35 | 0.51 | 0.71 | 1.00 |
| 0.5 | 0.00 | 0.00 | 0.02 | 0.04 | 0.07 | 0.13 | 0.20 | 0.31 | 0.46 | 0.68 | 1.00 |
| 0.6 | 0.00 | 0.00 | 0.01 | 0.03 | 0.06 | 0.10 | 0.16 | 0.26 | 0.40 | 0.62 | 1.00 |
| 0.7 | 0.00 | 0.00 | 0.01 | 0.02 | 0.04 | 0.08 | 0.13 | 0.20 | 0.33 | 0.55 | 1.00 |
| 0.8 | 0.00 | 0.00 | 0.01 | 0.01 | 0.03 | 0.05 | 0.09 | 0.14 | 0.25 | 0.45 | 1.00 |
| 0.9 | 0.00 | 0.00 | 0.00 | 0.01 | 0.01 | 0.03 | 0.04 | 0.08 | 0.14 | 0.29 | 1.00 |
| 1 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | |

# How does the dealer make money

- If buying, dealer will price his ask $A_t$ at time t greater than what he thinks the asset is worth $V_{t-1}^{D}$ but less than $V_H$
- If buying, dealer will LOSE money if an informed trader hits his quote
  - $\frac{\theta_{t-1}\alpha}{\theta_{t-1}\alpha+\frac{1-\alpha}{2}}(A_t - V_H)$
- Dealer will WIN money if an uninformed trader hits quote
  - $\frac{\frac{1-\alpha}{2}}{\theta_{t-1}\alpha+\frac{1-\alpha}{2}}(A_t - V_{t-1}^{D})$

# What quotes should we send

- Quotes are:
  - $B_t = V_{t-1}^D - \sigma \frac{(2\alpha\theta_{t-1}(1-\theta_{t-1}))}{1+\alpha(1-2\theta_{t-1})}$
  - $A_t = V_{t-1}^D + \sigma \frac{(2\alpha\theta_{t-1}(1-\theta_{t-1}))}{1+\alpha(2\theta_{t-1}-1)}$
- Intuitively, it looks like the below

| | Bid | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | theta | | | | | | | | | | |
| alpha | 0 | 0.1 | 0.2 | 0.3 | 0.4 | 0.5 | 0.6 | 0.7 | 0.8 | 0.9 | 1 |
| 0 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 |
| 0.1 | 0.00 | 0.02 | 0.03 | 0.04 | 0.05 | 0.05 | 0.05 | 0.04 | 0.03 | 0.02 | 0.00 |
| 0.2 | 0.00 | 0.03 | 0.06 | 0.08 | 0.09 | 0.10 | 0.10 | 0.09 | 0.07 | 0.04 | 0.00 |
| 0.3 | 0.00 | 0.04 | 0.08 | 0.11 | 0.14 | 0.15 | 0.15 | 0.14 | 0.12 | 0.07 | 0.00 |
| 0.4 | 0.00 | 0.05 | 0.10 | 0.14 | 0.18 | 0.20 | 0.21 | 0.20 | 0.17 | 0.11 | 0.00 |
| 0.5 | 0.00 | 0.06 | 0.12 | 0.18 | 0.22 | 0.25 | 0.27 | 0.26 | 0.23 | 0.15 | 0.00 |
| 0.6 | 0.00 | 0.07 | 0.14 | 0.20 | 0.26 | 0.30 | 0.33 | 0.33 | 0.30 | 0.21 | 0.00 |
| 0.7 | 0.00 | 0.08 | 0.16 | 0.23 | 0.29 | 0.35 | 0.39 | 0.41 | 0.39 | 0.29 | 0.00 |
| 0.8 | 0.00 | 0.09 | 0.17 | 0.25 | 0.33 | 0.40 | 0.46 | 0.49 | 0.49 | 0.40 | 0.00 |
| 0.9 | 0.00 | 0.09 | 0.19 | 0.28 | 0.37 | 0.45 | 0.53 | 0.59 | 0.63 | 0.58 | 0.00 |
| 1 | 0.00 | 0.10 | 0.20 | 0.30 | 0.40 | 0.50 | 0.60 | 0.70 | 0.80 | 0.90 | |

| Offer | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| theta | | | | | | | | | | |
| 0 | 0.1 | 0.2 | 0.3 | 0.4 | 0.5 | 0.6 | 0.7 | 0.8 | 0.9 | 1 |
| 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 |
| 0.00 | 0.02 | 0.03 | 0.04 | 0.05 | 0.05 | 0.05 | 0.04 | 0.03 | 0.02 | 0.00 |
| 0.00 | 0.04 | 0.07 | 0.09 | 0.10 | 0.10 | 0.09 | 0.08 | 0.06 | 0.03 | 0.00 |
| 0.00 | 0.07 | 0.12 | 0.14 | 0.15 | 0.15 | 0.14 | 0.11 | 0.08 | 0.04 | 0.00 |
| 0.00 | 0.11 | 0.17 | 0.20 | 0.21 | 0.20 | 0.18 | 0.14 | 0.10 | 0.05 | 0.00 |
| 0.00 | 0.15 | 0.23 | 0.26 | 0.27 | 0.25 | 0.22 | 0.18 | 0.12 | 0.06 | 0.00 |
| 0.00 | 0.21 | 0.30 | 0.33 | 0.33 | 0.30 | 0.26 | 0.20 | 0.14 | 0.07 | 0.00 |
| 0.00 | 0.29 | 0.39 | 0.41 | 0.39 | 0.35 | 0.29 | 0.23 | 0.16 | 0.08 | 0.00 |
| 0.00 | 0.40 | 0.49 | 0.49 | 0.46 | 0.40 | 0.33 | 0.25 | 0.17 | 0.09 | 0.00 |
| 0.00 | 0.58 | 0.63 | 0.59 | 0.53 | 0.45 | 0.37 | 0.28 | 0.19 | 0.09 | 0.00 |
| | 0.90 | 0.80 | 0.70 | 0.60 | 0.50 | 0.40 | 0.30 | 0.20 | 0.10 | 0.00 |

# Adding size to the model

- True value of security is uncertain $V \sim N(p_0, \sigma_v^2)$
- Two types of traders again:
  - Informed: Knows V and decides to buy/sell Q(V)
  - Uninformed: Doesn't know V and just buy/sells $X \sim N(0, \sigma_x^2)$
- We have a dealer again
  - Posts quotes p(O) which clear order imbalance $O = X + Q(V)$
  - Quotes are prices for trading all of O units

# Model Assumptions

- Dealer should make zero profit p(o)=E[V|O=o]
- Informed trader wants to maximize profits given p(o) by deciding how much to trade
- Dealer's pricing function is $p(o) = p_0 + \lambda * o$
- Informed trader: $Q^*(v) = \beta(v - p_0), \lambda = \frac{\sigma_v}{2*\sigma_x}, \beta = \frac{1}{2*\lambda}$
- Notes:
  - The more uncertain the price, the wider we quote
  - The higher the variance of uninformed trading, the tighter we quote
  - Informed trader also cares about the uninformed trader – the higher the variance of uninformed trading, the more he sizes up
- Empirically, we measure price impact as (used in hw)
  - $P_{t+1} - P_t = \lambda q_{t+1} + \epsilon_{t+1}$

# Homework 01 - Instructions

- For Stocks, use Databento for the data
  - The starter notebook will be shared over the weekend.
- For Currency, please use the two files in the Data Folder
  - We are still working to get access to Currency data from another provider, but for assignment 01, you can use the one given.

# Linking Results to Assignment 01

- Bid-Ask Spread
  - Expect spread to be wider when more adverse selection is present, namely $\alpha$ large
  - Expect spread to be wider when there is more uncertainty, namely $\sigma$ larger
  - Expect spread to be wider if information is less revealed, namely $\vartheta \approx \frac{1}{2}$
    - Can we link these to fundamentals
- Larger trades should be more informed

# Linking Results to Assignment 01

- Price Impact and Depth
  - They should be inversely related
    - Is their product a constant?
  - Low Price Impact/High Depth if there is relatively more noise trading/less informed trading
  - Informed traders act strategically and react to price impact
- Should spreads and depth be linked?

# Why?

- We want to gain an understanding how information revelation is happening
- Very quick information revelation would mean all price movements are permanent
  - Remember our models
- In this context what does return autocorrelation imply?
  - Is a high correlation corresponding to quick information revelation?
- Should the results differ for transactions/mid-quotes?

# Underlying Markets

**Stock Market**

- Multiple Trading venues
  - Exchanges, ATS, etc.
- Market Open 9:30-4:00pm
  - Opening/Closing Auctions
- Observe all trading

**EBS (Foreign Exchange Data)**

- One Trading Venue
- Continuously Open
  - Sydney Monday morning to New York Friday afternoon
- Observe only small volume
  - Only on EBS
  - More importantly only Dealer to dealer

# What is each Variable?

- Volume
  - Total $-amount of stocks traded
- No. of trades
- No. of orders
- VWAP (Volume weighted average price)
  - $VWAP = \sum_t P_t \cdot \frac{Q_t}{Total\ Quantity}$

# What is each Variable?

- BBO spread
  - Best Bid-Offer Spread = Bid-Ask Spread
  - Best Ask- Best Bid
  - Report in ct or bps
  - Be sure to specify calculation: Mean calculated time weighted/volume weighted…

# What is each Variable?

- BBO depth
  - No. of shares at Best Ask/Best Bid
  - Report depth at Ask and Bid separately
  - Calculate Time Weighted
- Depth at twice average spread
  - On Ask Side: Sum of all shares available at a price $P_t \leq Midquote_t + 2 \cdot Avg\ Spread$
  - Report Ask and Bid Separately
  - Again, calculate for each t and average across t

# What is each Variable?

- Price Impact
  - Regress $\frac{P_{t+5} - P_t}{P_t} = \lambda X_t + \epsilon_t$
  - $P_t$ are mid-quotes at time of trade
- How to get mid-quotes?
  - Match trade time to order book midquote
  - Likely hard to get exact pico-sec. match
  - Be sure to specify how match happened:
    - Is trade matched with closest available quote?
    - Trade matched to last available quote?
    - Trade matched to next available quote?
    - EXPLAIN CHOICE. THINK ABOUT WHAT CHOICE MEANS

# Data Required (Part a)

- Orderbook
  - Create order book at 1sec frequency
  - Calculate Midquote at each second
  - Use Midquote timeseries only
- Trade data
  - Use all trades (with relevant filter)
  - For each second, use the most current trade
  - Use most recent (at second level) time series

# Data Manipulation (Part b)

- Turn raw price data into returns data
  - Take log-prices $\log(p_t)$
  - Calculate log-returns $\mathrm{r_t} = \log(p_t) - \log(p_{t-1})$
  - We will work with log-returns data
  - Do this for transaction and mid-quotes

# Understand Data (Part c)

- Create summary statistics for the returns data
  - Should have a drift in the direction of the price change
  - How variable are returns?
    - Which time series is more variable? Is there a reason for this?
  - Compare to Assignment 1 (more on this later)

# Information Revelation (Part d)

- In python use *statsmodels.tsa.stattools.acf*
- For plotting the results use *statsmodels.graphics.tsaplots.plot_acf*
- Alternatively use *statsmodels.stats.diagnostic.acorr_ljungbox*
  - Only tests whether autocorrelations are significant
- Or other ACF function

# Arbitrage Opportunities (Part e)

- Now need order book for each currency
- At each second calculate the whether an arbitrage opportunity exists
  - Make sure to **buy at the Ask and sell at the bid**
- Duration: Seconds from start of opportunity to end of opportunity
- Frequency: Seconds with arbitrage opportunity
- Quantity: Minimum of available best bid/ask
  - Can extend this to multiple arbitrage levels, but it will become messy
- Calculate: $ to be made
- Set Quantity in relationship to general depth available

# Time Variation within the day

- Calculate variances, autocorrelation of returns in 30 minute window
- Show how they vary thoughout the day
- Comment on time variation
  - E.g. Does variance in returns comove with depth, price impact? Do autocorrelations move with liquidity?
  - Etc…

# Comments Homework 01

- **Spend more time on the interpretation of your results**
- If results seem off to you, say so
  – Are there reasons where our models abstracted from the real world?
- Please don’t send last minute questions ;)
- For currencies use at least of decimal points
  – EUR/USD 1.xxxx EUR/JPY 1xx.xx

# General "Expected" Findings HW01

- Spreads are larger in the morning and decreasing over the day for stocks
- Price Impact is positive (A market Buy order moves the price up)
- Spreads comove negatively with Depth
- More trading activity in larger cap stocks
- More trading activity means lower price impact, more depth (controlling for price), smaller spreads

# General “Expected” Findings HW01

- Spread pattern is not in currencies, but spreads are lower during market open
- Be careful on spread and volatility for currencies
  - Can get 1 Euro for approx. $1, so 0.0001 spread
  - Can get 1 AAPL share for approx. $600, so the same spread is 0.06
- Need to look at the results relative to prices

# Expected Results for Homework 01

- Important takeaways:
  - How to deal with an 5s Order Book
  - How to deal with Spread and Depth
  - Use Level 1 only - but Level 2 and 3 could also be incorporated
  - Differences between Trade and Midquote
  - Autocorrelations lags?
  - Price Impact?